Lab 10: ML - Unsupervised Learning (K-Means Clustering)

Objective:

- To use **K-Means clustering** to group similar Iris flowers based on their features (sepal and petal size).
- To learn how K-Means works without using labels, just based on the data.

Key Concepts:

- **K-Means Clustering** (Unsupervised Learning)
- **Elbow Method** for choosing the optimal number of clusters
- **Cluster Visualization** and analysis

Step 1: Import Required Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
```

Step 2: Load the Iris Dataset

```
# Load the Iris dataset
iris = load_iris()

# Convert the dataset into a DataFrame for easier manipulation
iris_data = pd.DataFrame(iris.data, columns=iris.feature_names)

# Show the first few rows of the dataset
iris_data.head()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

Step 3: Data Preprocessing

```
# The K-Means algorithm works better if all data is on the same scale.
# We will 'scale' the data to make sure everything is treated equally.
scaler = StandardScaler()
X_scaled = scaler.fit_transform(iris_data)
```

Step 4: Use the Elbow Method to Find the Best Number of Clusters

```
# Elbow method to find optimal k
wcss = []
for i in range(1, 11):  # Test k from 1 to 10
    kmeans = KMeans(n_clusters=i, init='k-means++', max_iter=300, n_init=10,
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Plotting the Elbow method
plt.plot(range(1, 11), wcss)
plt.grid(True) # Turns on the grid lines
plt.title('Elbow Method for Optimal k')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```

```
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```
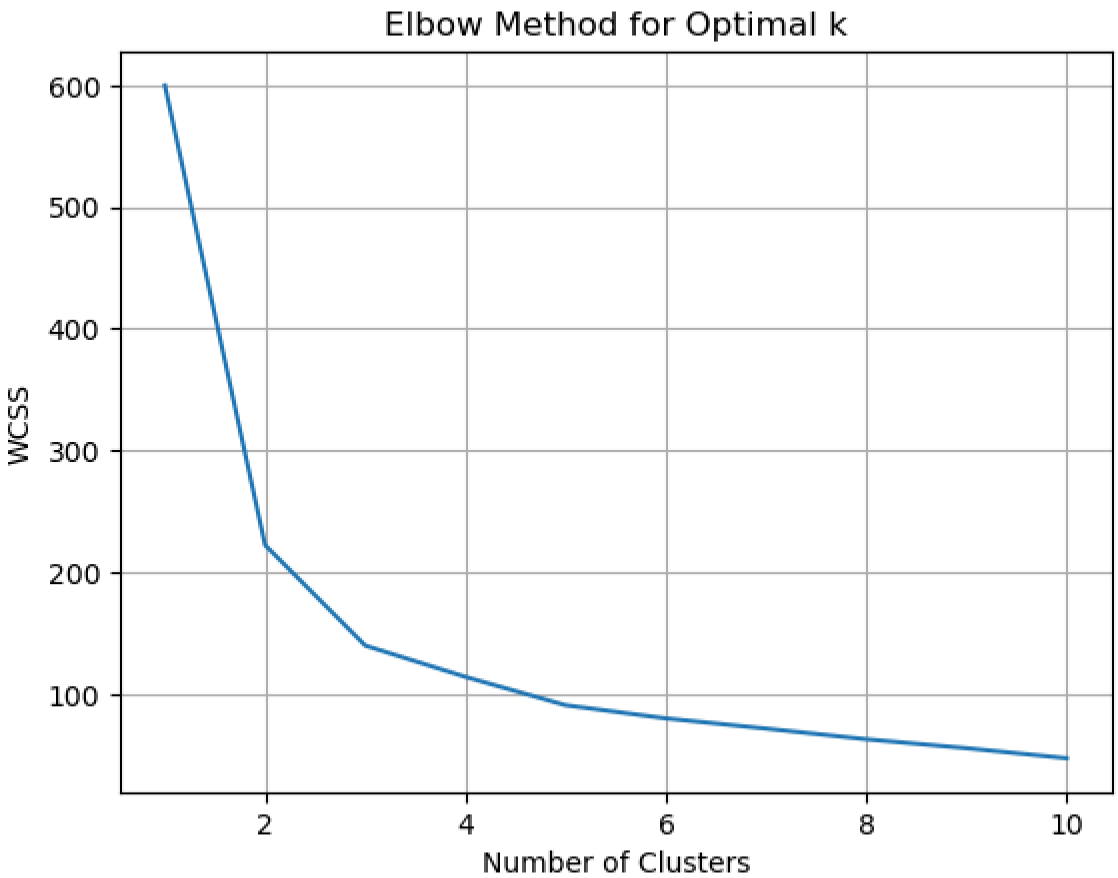



Step 5: Apply K-Means Clustering

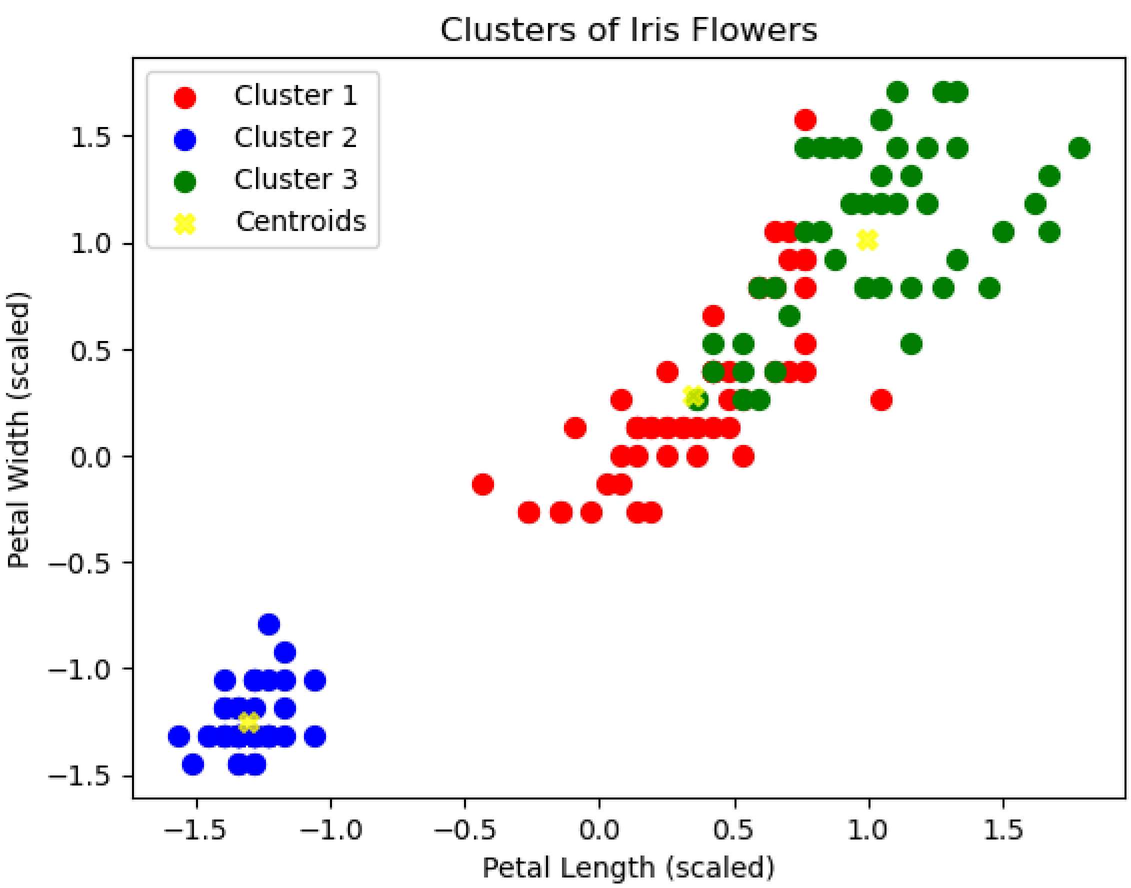
```
# After using the Elbow Method, we decide that k=3 is the best number of clu
# Now we apply K-Means with k=3 to cluster the flowers into 3 groups.
kmeans = KMeans(n_clusters=3, init='k-means++', max_iter=300, n_init=10, ran
```

```
y_kmeans = kmeans.fit_predict(X_scaled)
```

```
c:\Users\UNS\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: Use
  warnings.warn(
```

Step 6: Visualize the Clusters

```
# Let's visualize the clusters using Petal Length and Petal Width for simpli
plt.scatter(X_scaled[y_kmeans == 0, 2], X_scaled[y_kmeans == 0, 3], s=50, c=
plt.scatter(X_scaled[y_kmeans == 1, 2], X_scaled[y_kmeans == 1, 3], s=50, c=
plt.scatter(X_scaled[y_kmeans == 2, 2], X_scaled[y_kmeans == 2, 3], s=50, c=

# Plotting the cluster centers
plt.scatter(kmeans.cluster_centers_[:, 2], kmeans.cluster_centers_[:, 3], c=
            marker='X', label='Centroids')
plt.title('Clusters of Iris Flowers')
plt.xlabel('Petal Length (scaled)')
plt.ylabel('Petal Width (scaled)')
plt.legend()
plt.show()
```



Step 7: Analyze the Results

```
# Add the cluster labels to the original dataset
iris_data['Cluster'] = y_kmeans

# Show the average values for each feature per cluster
iris_data.groupby('Cluster').mean()
```

Cluster	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.801887	2.673585	4.369811	1.413208
1	5.006000	3.428000	1.462000	0.246000
2	6.780851	3.095745	5.510638	1.972340

Lab Tasks for Submission

Task 1 — Load the Iris Dataset & Select Two Fixed Features

You have to:

1. Load the Iris dataset from sklearn.datasets.
2. Extract ONLY these two features:
   - Sepal Length (cm)
   - Sepal Width (cm)
3. Create a pandas DataFrame with only these two columns.
4. Print the first 10 rows.

Task 2 — Apply K-Means Clustering (K = 3)

You have to:

1. Apply KMeans(n_clusters=3) using only Sepal Length and Sepal Width.
2. Add the cluster labels to the DataFrame.

Task 3 — Visualize the Clusters

You have to:

1. Create a scatter plot using:
   - X-axis → Sepal Length
   - Y-axis → Sepal Width